Phonetic Guided Speech Generation, Phase 1: Forced Alignment and Duration Prediction (ML Baseline)

Shreyash Golhani, Vaageesh Kumar Singh, Anwesha Adhikari, Ananya Narang, Aman Kumar
Advanced Machine Learning / Deep Learning Project, Rishihood University
Code: https://github.com/Shreyashgol/phonetic-guided-speech-generation

***Abstract*—Text-to-speech (TTS) needs to know *when* each phoneme occurs and *how long* it lasts. This report covers Phase 1 of a project comparing explicit phoneme-duration alignment with a learned acoustic generator. Phase 1 is the classical machine-learning part: (i) forced alignment of LJSpeech with the Montreal Forced Aligner (MFA) to obtain phoneme-level ground truth, (ii) exploratory analysis of the resulting durations, and (iii) a ladder of duration predictors from group-mean baselines to ridge regression, gradient-boosted trees and a BiLSTM. 0 of 13,100 utterances failed to align; 1,684 (12.9%) were removed by quality checks, mostly because of out-of-dictionary words. On a held-out, utterance-disjoint test set the best model (BiLSTM) reaches a mean absolute error of 1.330 frames (15.4 ms), against 2.400 frames for the per-phoneme-mean baseline; the improvement is 1.071 frames (95% bootstrap CI 1.048–1.092). We also analyse where the models fail and which feature groups matter.**

***Index Terms*—forced alignment, phoneme duration, text-to-speech, hidden Markov models, LJSpeech, Montreal Forced Aligner**

I. Introduction

A. Problem and scope

An explicit duration model tells a TTS system how many acoustic frames each phoneme occupies, but it does not produce timbre, coarticulation or prosody. The full project (Phases 1 and 2) asks whether a learned acoustic generator fixes what duration-only synthesis lacks. **Phase 1 covers only the ML component**: alignment, data analysis, feature engineering and duration regression. No audio is synthesised here.

B. Objectives

1) Produce reliable phoneme-level duration labels for LJSpeech and quantify their quality.
2) Characterise duration statistics and test them against the HMM duration model that underlies forced alignment.
3) Build and fairly compare duration predictors of increasing capacity.
4) Document failure modes and set up reusable assets for Phase 2.

C. Contributions

- A reproducible pipeline: text cleaning → MFA alignment → QA → features → models → evaluation, driven by one shared audio configuration.
- Alignment QA with four proxy checks that detect *bad* (not only failed) alignments.
- A frozen, utterance-level train/validation/test split to be reused unchanged in Phase 2.
- Utterance-bootstrap confidence intervals, a paired significance test and a feature-group ablation.

TABLE I
WHERE EACH PHASE 1 CRITERION IS ADDRESSED.

Criterion	Section	Evidence in repository
Literature review	II	references; MFA, FastSpeech, Grad-TTS
Dataset quality & EDA	III	notebooks 01–04; `alignment_qa.csv`
Feature engineering	V	notebook 05; `feature_sets.py`
Theoretical rigor	IV	derivations; HMM-vs-data test (nb. 04)
Model application	VI,VII	notebooks 06–08
Code quality	X	public repo, configs, frozen splits
Presentation, viva	XI	per-member contribution table

D. Evaluation map

Table I links each Phase 1 evaluation criterion to where it is addressed.

II. Literature Review

Forced alignment. Classical alignment models each phoneme as a left-to-right hidden Markov model (HMM) with Gaussian-mixture emissions over spectral features and finds the best state path by Viterbi decoding [4], [5]; parameters are trained with expectation–maximisation (EM) [6]. The Montreal Forced Aligner (MFA) packages this on top of Kaldi [7], training monophone then triphone models and offering pretrained English models and dictionaries [1]. Alignment quality depends on the pronunciation dictionary and transcript fidelity, so out-of-vocabulary words and noisy transcripts are the main failure sources.

Duration modelling in TTS. In HMM-based synthesis, duration is implicit in state self-transition probabilities, which yields geometric-like distributions that fit natural speech poorly [4]. Neural TTS made duration explicit: FastSpeech trains a duration predictor on alignments from a teacher and uses a length regulator to expand phoneme encodings to frame length [8]; FastSpeech 2 predicts log-domain durations directly

from forced-alignment targets [9]. Our Phase 1 predictors follow the same recipe (log-domain targets, alignment-derived labels).

Diffusion-based acoustic generation. Grad-TTS generates mel-spectrograms with a score-based diffusion decoder conditioned on text and a duration alignment [2]; a neural vocoder such as HiFi-GAN converts mels to waveforms [10]. This is the Phase 2 context; Phase 1 supplies its duration labels and baseline.

Gap addressed. Published systems rarely report how much duration is predictable from text alone, nor how alignment label noise limits the achievable error. We measure both on a single-speaker corpus.

III. Dataset, Alignment and EDA

A. Corpus

LJSpeech 1.1 [3]: 13,100 clips from one female speaker reading non-fiction (public domain), ≈23.92 h in total. Clip length: min 1.11 s, mean 6.57 s, max 10.10 s. All files are 22,050 Hz, mono, 16-bit PCM (checked on every file). The transcript file has raw and normalised text; 1,505 rows differ (numbers and abbreviations are expanded in the normalised column, which we use). After our cleaning (lowercase, no punctuation) the text contains 14,058 distinct words, of which 5,513 occur once.

B. Text preparation

For MFA we lowercase, strip accents (Unicode NFKD), replace hyphens by spaces and remove all punctuation. Punctuation is not discarded: each word keeps a code (none / comma-like / sentence-final) used later as a feature. A *shared phoneme inventory* (15 vowels, 24 consonants in ARPAbet, plus `sil` and `spn`) is defined once and imported by both phases.

C. Forced alignment

We use MFA 3.4.3 with the pretrained `english_us_arpa` acoustic model and dictionary, single-speaker mode, no retraining; aligning all 13,100 clips took about 22 minutes on a Colab CPU. Phone boundaries are converted from seconds to mel frames at 22,050 Hz, hop 256 (11.6 ms/frame) with *cumulative rounding*, so intervals remain contiguous. The shared audio configuration (FFT 1024, hop 256, 80 mels, 0–8 kHz) matches common pretrained vocoders.

Dictionary coverage. 420 word types (1,863 tokens, 0.83% of all tokens) are absent from the dictionary; MFA maps them to the unknown-noise phone `spn`. The most frequent are abbreviations and names (*mrs*, *mr*, *fbi*, *motorcade*, *connally*).

Failures. 0 of 13,100 utterances produced no alignment and are logged in `failed_utterances.csv`.

Quality control for bad alignments. No ground truth exists, so four proxy flags are computed per utterance: (i) any `spn` phone (1,562 utterances); (ii) speech rate (phones per second of speech) outside the 0.5th–99.5th percentile, i.e. 8.3–14.3 phones/s (132); (iii) more than 20% of speech phones at $\leq$2 frames (0, so this check never triggered); (iv) leading silence $>$ 1 s or trailing silence $>$ 1.5 s (8). The flags overlap; 1,684 utterances (12.9%) are excluded and 11,416 remain, dominated by the `spn` flag. Random and flagged clips were also inspected visually (boundaries on the mel spectrogram) and by listening.

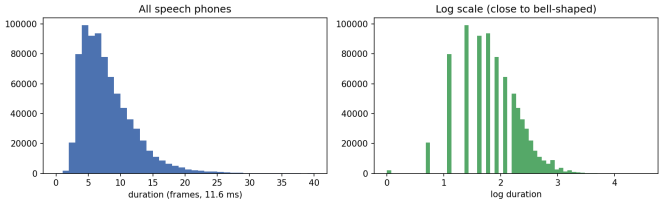


Fig. 1. Duration of all speech phones in frames (left) and on log scale (right).

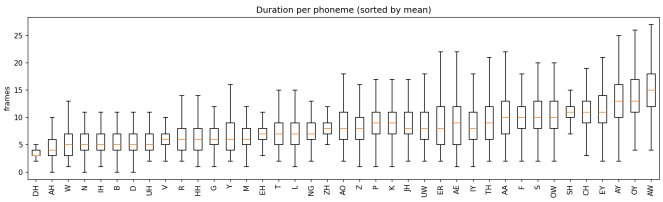


Fig. 2. Duration per phoneme, sorted by mean.

D. Exploratory analysis

The 777,055 retained speech phones have mean duration 7.46 frames (median 7, maximum 93). 293 phones rounded to zero frames and are clipped to one frame.

- **Skew.** Durations are right-skewed; $\log d$ is far more symmetric (Fig. 1), motivating log-domain regression.
- **Class.** Class means range from 5.78 frames (glides) and 6.07 (nasals) to 8.27 (fricatives) and 10.39 (affricates); vowels average 7.69 and stops 7.33, but vowels vary most (std 4.66 frames). Individual phones differ more than classes: DH averages 3.67 frames and AW 15.31 (Fig. 2).
- **Context.** Primary-stressed vowels are longer than unstressed ones by 3.56 frames (9.37 vs. 5.81); phones before a pause by 6.14 frames (13.28 vs. 7.14); word-final phones by only 0.59. Interior pauses have a median length of 15 frames. These effects define the feature groups of Sec. V.

IV. Theoretical Foundations

A. HMM forced alignment

Each phoneme is a 3-state left-to-right HMM with transition probabilities a_{ij} and emission density $b_j(\mathbf{o}) = \sum_m c_{jm}\,\mathcal{N}(\mathbf{o};\boldsymbol{\mu}_{jm},\boldsymbol{\Sigma}_{jm})$ over MFCC frames $\mathbf{o}_t$ (10 ms shift). Given the transcript, the phoneme HMMs are concatenated into one utterance HMM, so only the state path is unknown. Viterbi decoding finds

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i)\,a_{ij}\right] b_j(\mathbf{o}_t), \tag{1}$$

and back-pointers recover the best path; phoneme boundaries are the points where the path leaves one phoneme's states.

TABLE II
DATASET AND ALIGNMENT SUMMARY.

Item	Value
Clips (22,050 Hz, mono)	13,100
Total duration	23.92 h
Clip length: mean (min–max)	6.57 s (1.11–10.10)
Vocabulary after cleaning	14,058 words
Out-of-dictionary word types / tokens	420 / 1,863
Failed alignments	0
Excluded by quality checks	1,684 (12.9%)
Utterances kept	11,416
Speech phones kept	777,055
Mean phone duration	7.46 frames

Acoustic parameters are estimated by EM: the E-step uses forward–backward posteriors (soft state occupancies, summed over all paths), the M-step re-estimates means, covariances, mixture weights and transitions [4], [6]. *Viterbi answers "what is the single best path"; forward–backward answers "how likely is state s at frame t over all paths".*

B. Duration under an HMM

A state with self-loop probability a gives a geometric duration $P(d) = a^{d-1}(1-a)$ with mean $1/(1-a)$, whose mode is always 1. A phone of three left-to-right states has the sum of three such durations, a negative binomial shifted by 3:

$$P(d) = \binom{d-1}{2}(1-a)^3 a^{\,d-3}, \quad d \geq 3, \quad \mathbb{E}[d] = \frac{3}{1-a}. \tag{2}$$

Both have a fixed shape that cannot adapt to context; Sec. VII-D compares them to the data.

C. Regression on log-duration

Let $d_i \geq 1$ be the duration in frames of phone i and $y_i = \log d_i$. A predictor f is trained by (masked) squared error in log space,

$$\mathcal{L} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \big(f(\mathbf{x}_i) - y_i\big)^2, \qquad \hat{d}_i = \exp f(\mathbf{x}_i). \tag{3}$$

Because $\log$ is monotone, $\exp f$ estimates the conditional median when log-residuals are roughly symmetric, and the median minimises expected absolute error, our main metric.

D. Model classes

Ridge regression minimises $\|\mathbf{y} - X\mathbf{w}\|^2 + \alpha\|\mathbf{w}\|^2$; the L_2 penalty shrinks weights of rare one-hot categories [11]. **Gradient-boosted trees** add trees fitted to the residual gradient of the loss [12], capturing non-linear interactions (e.g. stress × pause). **BiLSTM** reads the whole phone sequence in both directions [13]; right context matters because phrase-final lengthening depends on what follows.

TABLE III
FEATURE GROUPS (IDENTICAL FOR ALL MODELS).

Group	Features	Motivation
identity	phone, stress, class, is-vowel	intrinsic duration
context	previous/next two phones	coarticulation
word	position and length in word, first/last	word-final lengthening
utterance	word index, relative position, lengths	phrase effects
pause/punct.	pause before/after, punctuation after word	phrase-final lengthening

V. FEATURE ENGINEERING

Unit and target. One row per speech phone; silence and `spn` rows are not modelled but signalled by `pause_before/after`. Target: duration in frames (clipped at 1). Table III lists the five feature groups.

Splitting and leakage control. Phones of one sentence are strongly correlated, so splitting by phone would leak. We split by *utterance* over all 13,100 ids with a fixed seed (42): train/val/test = 11,790/655/655 utterances (90/5/5%), stored as files and never regenerated. Alignment exclusions are applied afterwards, so the test set is stable. Hyper-parameters are tuned on validation only; the test set is scored once.

VI. MODELS AND EXPERIMENTAL PROTOCOL

A. Models

- **Global mean**: one value (floor).
- **Phone mean** (AML baseline): median duration per phoneme.
- **Back-off table**: median per (phone, stress, pause-after), backing off to (phone, stress) then phone when fewer than 20 training rows.
- **Ridge**: one-hot categoricals, standardised numerics, target $\log d$; $\alpha \in \{0.1, 1, 10, 100\}$.
- **Gradient-boosted trees** (histogram-based): categorical splits; (learning rate, iterations) $\in \{(0.1, 200), (0.05, 400)\}$.
- **BiLSTM**: embeddings (phone 64, stress 8, class 8, punctuation 4) and a linear layer on 11 numeric/flag inputs, 2-layer bidirectional LSTM (hidden 128, dropout 0.2), MLP $256 \to 64 \to 1$; 650,293 parameters. AdamW (lr 2×10^{-3}, weight decay 10^{-4}), batch 64, $\leq$30 epochs, learning rate halved on plateau, gradient clipping 1.0, early stopping (patience 5) on validation MAE; 25 epochs run; seed 0; trained on a Colab CUDA GPU.

B. Metrics

With $\hat{d}_i$ the prediction and N test phones: $\mathrm{MAE} = \frac{1}{N}\sum|\hat{d}_i - d_i|$, $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum(\hat{d}_i - d_i)^2}$, $R^2 = 1 - \sum(\hat{d}_i - d_i)^2 / \sum(d_i - \bar{d})^2$. Frames convert to ms by $\times 11.6$. **Confidence intervals**: 1000 bootstrap resamples of *utterances* (not phones, which are not independent); 95% percentile intervals. **Significance**: paired bootstrap of the MAE difference between baseline and best model on identical resamples.

TABLE IV
DURATION PREDICTION ON THE TEST SET (FRAMES OF 11.6 MS; 95% CI BY UTTERANCE BOOTSTRAP).

Model	MAE	95% CI	MAE (ms)	RMSE	R^2
BiLSTM	**1.330**	[1.310, 1.353]	15.4	1.949	0.775
Gradient boosting	1.540	[1.517, 1.566]	17.9	2.265	0.697
Ridge	2.013	[1.987, 2.043]	23.4	2.877	0.511
Back-off table	2.220	[2.193, 2.248]	25.8	3.122	0.424
Phone mean (baseline)	2.400	[2.371, 2.431]	27.9	3.515	0.269
Global mean	3.138	[3.110, 3.167]	36.4	4.112	0.000

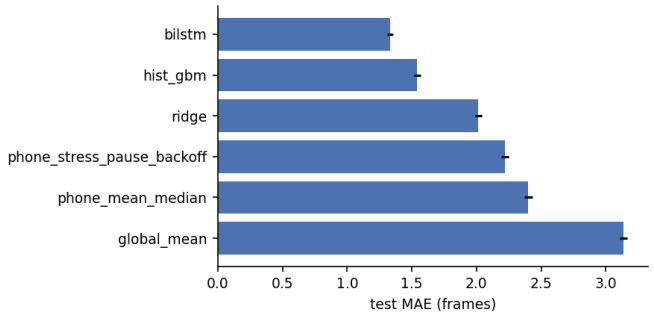


Fig. 3. Test MAE with 95% utterance-bootstrap intervals.

TABLE V
FEATURE-GROUP ABLATION (GRADIENT-BOOSTED TREES, TEST SET, FRAMES).

Setting	MAE	RMSE	R^2	ΔMAE
All features	1.540	2.265	0.697	0.000
Without identity	2.099	3.011	0.464	+0.559
Without context	1.942	2.736	0.557	+0.402
Without word	1.601	2.349	0.674	+0.061
Without utterance	1.534	2.258	0.699	−0.006
Without pause/punctuation	1.654	2.541	0.618	+0.114
Identity only	2.388	3.472	0.287	+0.848

VII. RESULTS

A. Model comparison

Table IV shows the test results. The best model, BiLSTM, has MAE 1.330 frames (15.4 ms; CI 1.310–1.353) and $R^2 = 0.775$. Relative to the phone-mean baseline (2.400 frames, 27.9 ms), the paired MAE gain is 1.071 frames (95% CI 1.048–1.092). The ranking is monotone in model capacity: global mean 3.138 > phone mean 2.400 > back-off 2.220 > Ridge 2.013 > gradient boosting 1.540 > BiLSTM 1.330 frames. The BiLSTM beats gradient boosting by 0.21 frames (1.330 vs. 1.540; R^2 0.775 vs. 0.697) and the 95% intervals ([1.310, 1.353] vs. [1.517, 1.566]) do not overlap. Ridge's validation MAE was insensitive to α (2.021–2.022), so its limit is linearity, not regularisation.

B. Ablation

Removing the identity features (phone, stress, class) hurts most (Table V). Context features come next (+0.402 frames), then pause/punctuation (+0.114) and word features (+0.061); utterance-level features give no measurable benefit (−0.006). Identity features alone reach 2.388 frames, about the phone-mean baseline (2.400), so context, pause and word information account for 0.85 frames of improvement. This matches Sec. III: the pre-pause effect (+6.14 frames) is large, and only 5% of phones precede a pause, yet `pause_after` still matters because those phones carry the largest errors (error analysis below).

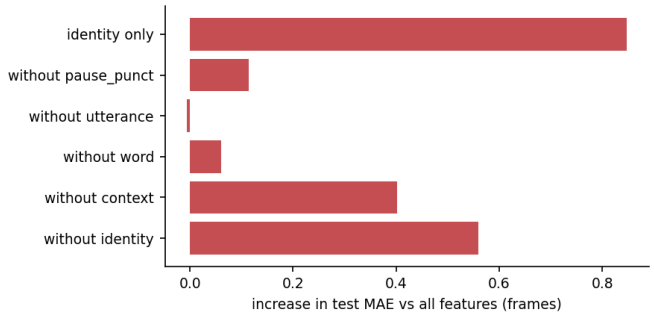


Fig. 4. Increase in test MAE when a feature group is removed.

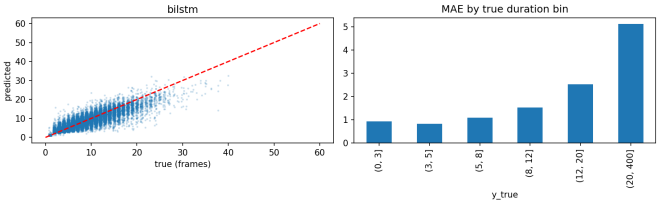


Fig. 5. Best model: predicted vs. true duration (left) and MAE by true-duration bin (right).

C. Error analysis

The largest relative error is in glides (relative MAE 20.4%) (Fig. 6). Errors show *shrinkage toward the mean*: phones of ≤ 3 frames are over-predicted by 0.88 frames on average, while phones of 12–20 and > 20 frames are under-predicted by 1.77 and 4.79 frames (MAE 2.52 and 5.13). Phones followed by a pause have MAE 2.64 against 1.26 otherwise. By absolute error the diphthongs and long vowels are worst (OY 2.21, AY 2.15, AE 2.11, AW 2.05 frames); by relative error glides (0.204), nasals (0.199) and liquids (0.193) are worst, with vowels at 0.184. The worst utterances (LJ024-0071, LJ017-0250, LJ036-0073) have mean errors of 3.17, 2.33 and 2.10 frames against 1.33 overall; Six of the ten worst utterances come from only two chapters: LJ028 (four; archaic wording with names and "thou") and LJ024 (two; the text of a political speech). This suggests that unusual vocabulary and oratorical delivery, rather than random clips, drive the largest errors; we did not isolate individual alignment mistakes.

D. HMM duration model versus real durations

Over the 12 most frequent phones the 3-state negative-binomial model fits *better* only for longer phones (S −2.87 vs. −3.35 for geometric, Z −2.66 vs. −3.13, K −2.62 vs. −3.15; means 9–11 frames) and *worse* for short ones (AH −3.94 vs.

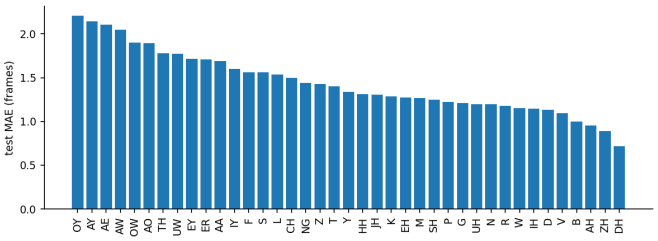


Fig. 6. Test MAE per phoneme (best model).

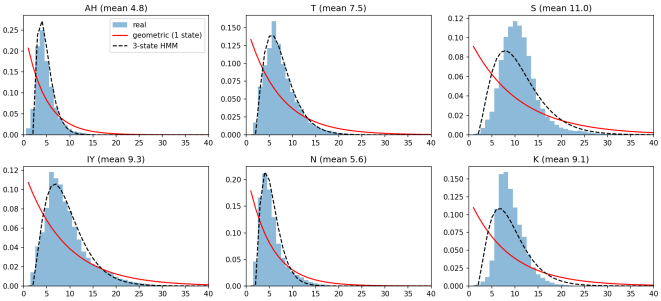


Fig. 7. Real duration histograms (bars) versus a geometric (single-state) and a 3-state negative-binomial model with the same mean.

-2.47, DH -4.64 vs. -2.15; also N, D, R), because durations below 3 frames are impossible under it yet occur for 12.4% of DH and 7.8% of AH. Neither fixed shape fits every phone, which motivates a context-aware learned predictor. Caveat: MFA works in 10 ms frames while we count 11.6 ms frames, so the comparison is approximate.

VIII. Discussion and Limitations

- **Label noise.** Targets come from forced alignment, whose boundaries are themselves HMM estimates on a 10 ms grid; our predictors cannot beat the alignment's own error, and some bias of the HMM duration model is inherited by the labels.
- **Irreducible variance.** Speaker choices (emphasis, rate within a sentence) are not recoverable from text, so R^2 has a ceiling below 1.
- **Quantisation.** Converting 10 ms alignment frames to 11.6 ms mel frames adds rounding of up to half a frame.
- **Single speaker, English, read speech.** Results may not transfer to expressive or multi-speaker data.
- **Exclusions.** Removing 12.9% of utterances (mostly `spn` from abbreviations and proper names such as *mr*, *mrs*, *fbi*) biases the retained set away from such text; extending the dictionary would recover most of them.
- **No audio yet.** WER via a held-out ASR and the listening study require synthesis and belong to Phase 2.

IX. Conclusion and Phase 2 Outlook

Phase 1 delivers validated duration labels (11,416 utterances), a frozen split and a documented comparison of duration predictors (best: BiLSTM, MAE 1.330 frames). Phase 2 will feed phonemes and (true or predicted) durations through a length regulator into a learned mel decoder (optionally diffusion-based, following Grad-TTS), render audio with a pretrained vocoder, and compare against duration-only synthesis using held-out-ASR WER, duration MAE and a small human intelligibility preference study, always on the same test split.

X. Reproducibility

Public repository: https://github.com/Shreyashgol/phonetic-guided-speech-generation. Order of execution: notebooks 00 $\rightarrow$ 09 (see `README.md`). Shared settings live in `configs/data.yaml` (22,050 Hz, FFT 1024, hop 256, 80 mels); splits in `data/splits/`; every figure and table of this report is regenerated by notebook 09 from files in `results/`. Seeds: split 42, models 0. Software: Python, PyTorch [14], scikit-learn [15], MFA 3.4.3. The dataset and checkpoints are not committed.

XI. Individual Contributions

Each member owns one area and can explain every stage in the individual viva (Table VI).

TABLE VI
Contributions.

Member	Main area	Notebooks / sections
Shreyash Golhani	Dataset exploration, EDA, theoretical rigor (HMM alignment, duration theory)	Notebooks 01–04; Sec. III, IV
Vaageesh Kumar Singh	Model application and rigor	Notebooks 06–08; Sec. VI, VII
Anwesha Adhikari	Model application and rigor	Notebooks 06–08; Sec. VI, VII
Ananya Narang	Feature engineering, theoretical rigor (HMM alignment, duration theory)	Notebook 04–05; Sec. V
Aman Kumar	Report, presentation, literature review, README	Sec. II, introduction, discussion, conclusion

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